Abhinav Kumar Singh

Cloud Data Engineer | Azure Databricks | Snowflake | PySpark | Azure Data Factory

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Professional Summary

Cloud Data Engineer with 2+ years of experience building and optimizing enterprise data pipelines on Azure Databricks, Snowflake, PySpark, SQL, and Azure Data Factory. Proven track record of processing 10M+ records across 1,000+ files, implementing Medallion Architecture, building reusable SFTP-to-ADLS file-transfer pipelines, and orchestrating production workflows. Strong in distributed data processing, ETL automation, data security, and performance optimization.

Technical Skills

Data Engineering: ETL/ELT Pipelines, Medallion Architecture, Batch Processing, SFTP File Transfer, Data Transformation, Data Quality, Data Masking, Encryption

Languages & Processing: Python, SQL, Spark SQL, PySpark, Apache Spark

Cloud & Data Platforms: Azure Databricks, Azure Data Factory, Azure Data Lake Storage (ADLS), Snowflake

Orchestration & Tools: Databricks Workflows, Control-M, Azure DevOps, CI/CD, Git, SSMS, Workflow Monitoring

Professional Experience

Tata Consultancy Services (TCS) **June 2024 – Present**

Cloud Data Engineer *Kolkata, West Bengal, India*

DFLE Migration | Azure Databricks, Snowflake, Azure Data Factory, Control-M

- Built and maintained production-grade ETL pipelines on Azure Databricks and Snowflake, processing **10M+ records across 1,000+ files** to power enterprise reporting and analytics.
- Implemented a Medallion Architecture (Bronze, Silver, Gold) in Databricks, improving query performance by **35%** and data maintainability.
- Optimized Spark SQL transformations and execution strategies, reducing processing time by **35%** and improving compute utilization.
- Orchestrated scheduled workflows with Azure Data Factory and Control-M, automating processing of 1,000+ files and supporting SLA adherence.
- Applied data-masking and encryption controls to protect **10M+ sensitive records**, strengthening data protection and compliance in production.
- Automated dependency detection and parameter extraction across ETL workflows, cutting manual operational effort by **30%**.

DataLynx | Azure Data Factory, SFTP, Azure Data Lake Storage (ADLS)

- Designed a reusable, parameter-driven generic pipeline that connects to multiple SFTP servers via configurable connection details, eliminating the need for server-specific pipelines.
- Configured a reusable linked service for secure inbound and outbound file transfers between SFTP servers and ADLS.
- Developed a parent orchestration pipeline that invokes the generic pipeline, moves files for processing, applies customer-defined filtering rules, and loads results into new business-aligned tables.

Projects

Walgreens Healthcare Data Platform *Azure Databricks, Spark SQL, Snowflake*

- Built scalable ETL pipelines to ingest structured and semi-structured healthcare data into Snowflake, supporting analytics across millions of records.
- Improved warehouse query performance and data availability through SQL tuning and query optimization.

E-Commerce Store Data Pipeline *Azure Databricks, Azure Data Factory*

- Designed a Medallion Architecture (Bronze, Silver, Gold) to improve data quality and analytical query performance.
- Automated data processing and reporting workflows, improving consistency and reliability of business reports.

Education

Chandigarh University **August 2020 – June 2024**

Bachelor of Engineering in Computer Science | CGPA: 7.98 *Punjab, India*

Certifications

- **Databricks Certified Data Engineer Associate** | Verify Credential

Achievements

- **Applause Awards (2x):** Recognized for timely, high-quality delivery on projects for a major pharmaceutical client using Databricks, Snowflake, and Azure Data Factory.
- **Spot Award (2x):** Recognized for high-quality data solutions on a pharmaceutical project using Azure Databricks, Snowflake, and Azure Data Factory.